Shota Suzuki

Ph.D. Student in Electrical Engineering and Information Systems | The University of Tokyo
Email (via Web) | Portfolio Website | Google Scholar | LinkedIn

RESEARCH INTERESTS & EXPERTISE

I am conducting research on novel memory-centric computing systems, tackling hardware-software co-design challenges by leveraging knowledge across algorithms, computer architecture, circuits, and semiconductor devices.

Areas of Expertise: Large Language Models (LLMs), Computer Architecture, Computation-in-Memory (CiM), Resistive Random-Access Memory (ReRAM)

EDUCATION

- **The University of Tokyo** | Tokyo, Japan
  Doctor of Engineering in Electrical Engineering and Information Systems (Apr 2026 – Present)
    - Advisor: Prof. Ken Takeuchi
    - Project: Advanced Human Resource Development Leading Green Transformation (SPRING GX)
- **The University of Tokyo** | Tokyo, Japan
  Master of Engineering in Electrical Engineering and Information Systems (Apr 2024 – Mar 2026)
    - Advisor: Prof. Ken Takeuchi
    - Master's Thesis: *"Research on Analysis of Changes in Current by Read-disturb on Multi-level Cell ReRAM and ReRAM Application for KV Cache of Large Language Models"*
- **The University of Tokyo** | Tokyo, Japan
  Bachelor of Engineering in Electrical and Electronic Engineering (Apr 2020 – Mar 2024)
    - Advisors: Prof. Shuichi Sakai and Prof. Hidetsugu Irie
    - Graduation Thesis: *"Microarchitecture Design of Dynamic Approximate Architecture Based on Execution Time"*

GRANTS & SCHOLARSHIPS

- **Supporting Pioneering Research through AI for 1,000 Discovery challenges Program (SPReAD)**, MEXT (Jul 2026 – Jan 2027)
    - Research Project: "Creation of Highly Power-Efficient AI Hardware: Agent-Driven Autonomous Processing-in-Memory Design"
    - Role: Principal Investigator (PI)
    - Awarded a highly competitive research grant aimed at driving scientific innovation through advanced AI technologies.
- **Research Fellowship / Grant (SPRING GX)**, JST SPRING Program, The University of Tokyo (Apr 2026 – Mar 2029)
- **Student Supporters Club Scholarship (Grant-type)**, The University of Tokyo Foundation (Apr 2024 – Mar 2026)

AWARDS

- **Award in the Large Language Model (LLM) Category**, *Chipathon (Semiconductor Design Hackathon)*, Graduate School of Engineering, The University of Tokyo (Feb 2026)

Developed and optimized a hardware-accelerated system to execute a language model on an FPGA board.

- **Best Research Award**, *The 8th cross-disciplinary Workshop on Computing Systems, Infrastructures, and Programming (xSIG 2024)* (Aug 2024)

RESEARCH & WORK EXPERIENCE

- **TOWN, Inc.** | *Part-time Engineer* (Oct 2022 – Sep 2025)
    - Built and operated cloud infrastructure using Amazon Web Services (AWS).
    - Automated infrastructure testing using Serverspec.
    - Participated in an R&D project to automate server configuration and construction using Microsoft Azure OpenAI.
- **University of Leoben** | Leoben, Austria
  Research Intern (IAESTE Internship Program) (Aug 2024 – Sep 2024)
    - Conducted research on the analysis of X-ray diffraction spectra using neural networks under the mentorship of Assoc. Prof. Ernst Gamsjäger.
- **The University of Tokyo |** *Teaching Assistant (TA)* (Mar 2022 – Jul 2025)
  Assisted and coordinated multiple laboratory and undergraduate courses:
    - *“Prototyping Experience to Shape Ideas - Prototype Design in the AI Era”* (Mar 2025 – Jul 2025)
    - *“Approximate Computing for Machine Learning Applications”* (Oct 2024 – Dec 2024)
    - *“Prototyping Experience to Shape Ideas - From Robots to Home Appliances”* (Multiple terms, served as TA Coordinator in 2023)
    - *“Analog Circuits”* (May 2024)
    - *“VLSI Architecture”* (Mar 2024 – May 2024)

CERTIFICATES & PROFESSIONAL TRAINING

- **Deep Learning Advanced Course: Deep Generative Models**, Matsuo-Iwasawa Lab, The University of Tokyo (Apr 2026)
  Advanced practical program on theories and implementation of VAEs, GANs, Energy-Based Models, and Diffusion Models.
- **AI and Semiconductor Course**, Systems Design Lab & Matsuo-Iwasawa Lab, The University of Tokyo (Oct 2025)
  Comprehensive training in semiconductor technologies, architecture, and hardware/software co-design for AI workloads.
- **Deep Learning Basic Course**, Matsuo-Iwasawa Lab, The University of Tokyo (Jul 2025)
  Practical program on fundamental deep learning theories and frameworks (PyTorch, CNNs).
- **Global Consumer Intelligence (GCI) Program**, Dept. of Technology Management for Innovation, The University of Tokyo (Feb 2024)
  Data science and machine learning training (data analysis and predictive modeling using Python).
- **Go Global Gateway (GGG) Certificate of Recognition**, Center for Global Education, The University of Tokyo (Nov 2023)
  Recognized for international engagement (experiential learning at KTH Sweden, study abroad at University of Hawaii, and UTokyo GUC Student Ambassador).

PUBLICATIONS

Refereed Journal Articles

1. Chihiro Matsui, Ayumu Yamada, **Shota Suzuki**, Naoko Misawa, and Ken Takeuchi, “NanoCiM: Nano-ReRAM Computation-in-Memory with Neural Network Weight-aware X-Address Scrambling to Suppress IR Drop-oriented Bit-line MAC Current Variation,” *IEEE Transactions on Circuits and Systems for Artificial Intelligence (TCASAI)*, vol. 3, no. 3, pp. 190-202, June 2026.
2. **Shota Suzuki**, Naoko Misawa, Chihiro Matsui, and Ken Takeuchi, “Mapping strategy of quantized KV cache of LLMs into mixture of SLC and MLC ReRAM among various quantization methods,” *Japanese Journal of Applied Physics (JJAP)*, vol. 65, pp. 03SP09, February 9, 2026.
3. Yuya Degawa, **Shota Suzuki**, Junichiro Kadomoto, Hidetsugu Irie, and Shuichi Sakai, “Cycle-Oriented Dynamic Approximation: Architectural Framework to Meet Performance Requirements,” *IEEE Computer Architecture Letters (CAL)*, vol. 23, no. 2, pp. 211-214, July-December 2024.

Refereed International Conferences

1. **Shota Suzuki**, Naoko Misawa, Chihiro Matsui, and Ken Takeuchi, “Hybrid SLC/MLC ReRAM for Quantized KV Cache in Transformer-based LLM,” *International Conference on Solid State Devices and Materials (SSDM)* Poster Late News, September 17, 2025, pp. 755-756.
2. **Shota Suzuki**, Naoko Misawa, Chihiro Matsui, and Ken Takeuchi, “Observation of State Change in 40nm TaO_X ReRAM Cells during Read-disturb,” *IEEE Silicon Nanoelectronics Workshop (SNW)*, June 8, 2025, pp. 26-27.

Refereed Domestic Conferences

1. **Shota Suzuki**, Yuya Degawa, Tomohiro Yoshita, Junichiro Kadomoto, Hidetsugu Irie, and Shuichi Sakai, “RTL Design and Soft Processor Implementation of an Architecture for Cycle-oriented Dynamic Approximation,” *The 8th cross-disciplinary Workshop on Computing Systems, Infrastructures, and Programming (xSIG 2024)*, August 7, 2024, pp. 1-14. **(Best Research Award)**

Non-Refereed Domestic Conferences / Workshops

1. **Shota Suzuki**, Naoko Misawa, Chihiro Matsui, and Ken Takeuchi, “Investigation of Current Fluctuations and RTN Frequency Change Induced by Read Stress in ReRAM,” *The 73rd JSAP Spring Meeting*, March 15, 2026.
2. **Shota Suzuki**, Naoko Misawa, Chihiro Matsui, and Ken Takeuchi, “ReRAM Mapping Method Based on Bit-Level Error Robustness of Quantized KV Cache,” *The 73rd JSAP Spring Meeting*, March 15, 2026.
3. **Shota Suzuki**, Naoko Misawa, Chihiro Matsui, and Ken Takeuchi, “Investigation of ReRAM Mapping Techniques for Storing Quantized KV Cache in LLMs,” *IEICE Technical Report*, vol. 125, no. 260, VLD2025-59, pp. 227-232, December 3, 2025.
4. **Shota Suzuki**, Naoko Misawa, Chihiro Matsui, and Ken Takeuchi, “Changes in Read Current and RTN Frequency due to Read-disturb in 40nm TaO_X ReRAM,” *IEICE Technical Report*, vol. 125, no. 261, ICD2025-50, pp. 74-79, December 2, 2025.
5. Daichi Mikubo, **Shota Suzuki**, Shu Sugita, Junichiro Kadomoto, and Hidetsugu Irie, “Acceleration of speech feature extraction using an architecture with dynamically controllable computational precision,” *The 24th Forum on Information Technology (FIT 2025)*, September 4, 2025.
6. Yifan Wang, Adil Padiyal, Daqi Lin, **Shota Suzuki**, Chihiro Matsui, and Ken Takeuchi, “A Design for Digital-CiM INT8 Transformer Accelerator with Pipeline,” *The 72nd JSAP Spring Meeting*, March 14, 2025.

SKILLS & INTERESTS

- **Languages:** Japanese (Native), English (Proficient), German (Duolingo 800+ days streak)
- **Interests & Hobbies:** Alpine Skiing (SAJ Level 2), Language Learning